Chapter 4: Photosynthesis

Life Science, Grade 7 · Unit 2: How Living Things Get Energy

4.1 Why plants do not need to eat

Animals, including people, must eat food to get energy. Plants are different: they make their own food. The process plants use to make food is called **photosynthesis**. The word comes from two Greek words: *photo*, meaning "light", and *synthesis*, meaning "putting together". In photosynthesis, a plant uses the energy of light to put together simple materials into sugar.

Photosynthesis happens mainly in the leaves. Leaf cells contain small green structures called **chloroplasts**. Chloroplasts contain a green pigment, **chlorophyll**, which absorbs light energy. Chlorophyll is the reason most plants look green.

4.2 What goes in and what comes out

To make sugar, a plant needs three things. It takes in **carbon dioxide** from the air through tiny openings on the underside of its leaves, called **stomata**. It takes in **water** from the soil through its roots; the water travels up the stem to the leaves. And it needs **light energy**, usually from the Sun.

Inputs (what the plant uses)	Where it comes from	Outputs (what the plant makes)	What happens to it
Carbon dioxide (CO_2)	Air, through the stomata	Glucose, a sugar ($C_6H_{12}O_6$)	Used for energy and to build the plant's body; extra is stored as starch
Water (H_2O)	Soil, through the roots	Oxygen (O_2)	Released into the air through the stomata
Light energy	The Sun	—	Stored as chemical energy in glucose

Scientists summarize photosynthesis with this chemical equation:

$$6CO_2 + 6H_2O \xrightarrow{\text{light}} C_6H_{12}O_6 + 6O_2$$

Read aloud, it says: six molecules of carbon dioxide and six molecules of water, using light energy, produce one molecule of glucose and six molecules of oxygen. Notice that the number of each kind of atom is the same on both sides of the arrow: atoms are not created or destroyed, only rearranged.

4.3 Why photosynthesis matters to you

Almost every food chain on Earth starts with photosynthesis. When a rabbit eats grass, it gets the energy that the grass stored in glucose. When a fox eats the rabbit, some of that same energy passes on again. Without plants and other organisms that photosynthesize, animals would have no source of food energy.

Photosynthesis also gives us the oxygen we breathe. Before photosynthesizing organisms appeared, Earth's air had almost no oxygen. Today about 21% of the air is oxygen, and photosynthesis on land and in the oceans keeps replacing the oxygen that animals use.

4.4 What limits photosynthesis

A plant cannot photosynthesize faster and faster forever. The rate depends on the amount of light, the amount of carbon dioxide, and the temperature. If any one of these is in short supply, it slows the whole process down, even if the others are plentiful. This is why greenhouse farmers sometimes add extra carbon dioxide to the air inside their greenhouses on sunny days.

At night there is no light, so photosynthesis stops. The plant keeps living by using the glucose it stored during the day.

Check your understanding

1. Name the three inputs of photosynthesis and where each one enters the plant.
2. A plant is kept in a dark cupboard for two weeks. Use the equation to explain what will happen to it.
3. Why do we say that the energy in your breakfast came from the Sun?